

Soybean — Canonical KB vs. Delta KB (post-fix run)

10 separate diseases (5 + 5, disjoint) | Post-fix half of phase0r_traces (1).jsonl | Soybean | Generated 2026-05-16

Context

After the consolidator token-budget fix the swarm re-ran on Nova. This report uses **only the post-fix half** of the trace file (76 clean Soybean records; the first 147 records are the stale pre-fix run and are excluded). It covers **10 separate Soybean diseases** — 5 in Table A and 5 *different* ones in Table B (the tables are disjoint).

Validation basis. Every `canonical_says` string is itself sourced (Phase-1 KB cites Crop Protection Network / university extension). A delta matching canonical is validated against that source; a delta that diverges is classified in Table B for human adjudication.

Table A — canonical present AND delta has a value (grounded)

5 distinct Soybean diseases. Pre-fix, 92% of deltas had `canonical_says = "(not specified)"`; these show grounding now works.

Disease	Field	Canonical KB said (<code>canonical_says</code>)	Delta KB value (<code>image_shows</code>)	Raw caption (<code>image_quote</code>)
Frogeye Leaf Spot	leaf_necrosis	Leaf lesions are small, irregular to circular, gray with reddish-brown borders, occurring most commonly on the upper leaf surface;	The image shows multiple small, irregular, and gray lesions with reddish-brown borders on the leaf.	Irregular shapes with reddish-brown borders are present.
Brown Stem Rot	stem_pith	white in healthy stems; brown in BSR	brown pith in the stem.	The photo shows a split stem with brown pith.
Soybean Rust	leaf_necrosis	Symptoms begin on leaves in the lower plant canopy with gray-green, tan to dark-brown, or reddish-brown angular lesions (2-5 mm diameter) developing first on the underside of leaves; small pustules form in the lesions and break open to release masses of tan spores; lesions may also appear on petioles, pods, and stems.	Brown spots are visible along the veins, indicating necrotic tissue extending inward.	Brown spots are visible along the veins, indicating necrotic tissue extending inward.
Root And Stem Rot	severity_visible	A dark chocolate-brown lesion on the lower stem extending from below the soil line upward from the taproot, often reaching several nodes and girdling the stem;	Dark brown discoloration extending several nodes up the stem.	The stem exhibits dark brown discoloration with a rough texture, indicating cankers.

Disease	Field	Canonical KB said (<code>canonical_says</code>)	Delta KB value (<code>image_shows</code>)	Raw caption (<code>image_quote</code>)
Septoria Leaf Spot	leaf_necrosis	Initial symptoms are small dark brown spots (less than 1/8 inch) on lower leaves that enlarge and grow together into irregular brown areas, often associated with yellow patches concentrated more on one side of the leaf than another;	Brown necrotic areas on the leaves, which are irregularly shaped and cover various parts of the leaf.	There are brown necrotic areas covering the leaves.

Table B — proper changes (delta extends / refines / contradicts canonical)

5 *different* Soybean diseases (no overlap with Table A). The right-hand column classifies the change.

Disease	Canonical KB	Delta KB (image)	The proper change
Charcoal Rot	brown to dark spots appear on cotyledons at seedling stage, reddish-brown to black lesions on unifoliate leaves, and light brown to grey superficial lesions on stems mid to late season, with reddish- brown discoloration in the pith and vascular tissues.	The stem pith is white with brown outer vascular tissue.	CONTRADICTION/RELOCATION — image puts discoloration in the vascular ring, not the pith
Bacterial Blight	Septoria brown spot (Septoria glycines)	The lesions are irregular, with a chlorotic yellow halo and scattered necrotic tissue, differentiating it from Septoria brown spot.	VALUE-ADD — supplies the discriminative criteria vs Septoria the canonical only named
Bacterial Pustule Of Soybean Disease	Bacterial pustule lesions lack the opening on top and spores that mature soybean rust pustules have; if an opening is present, it is typically a linear crack across the surface of the pustule (visible only under 20X magnification). Unlike bacterial blight, bacterial pustule lesions do not appear water soaked and will have raised centers.	The lesions are irregular, with some showing irregularities around the perimeter, and the lesion centers are tan, the margins are chocolate- brown, and there is a chlorotic yellow halo.	NEW FIELD-VISIBLE SIGN — adds a macroscopic diagnostic; canonical needed 20X magnification
Cercospora Blight	Light purple spots and areas develop on the top surface of leaves and expand to reddish-purple or bronze; infected leaves appear leathery and ‘sunburned,’ with red-brown spots that may coalesce into large necrotic areas leading to defoliation.	Dark brownish-green spots with a leathery texture on the leaf surface.	STAGE/COLOR REFINEMENT — captures a brownish-green phase, not canonical’s purple
Rhizoctonia Damping-Off, Blight And Rot	rusty-brown	The affected roots have brownish lesions with greenish areas.	ADDED OBSERVATION — notes greenish areas beyond canonical’s single ‘rusty-brown’

Verdict

- **Table A** — the fix restored *grounding*: the swarm reads the canonical slice and records confirmations against it across 5 separate Soybean diseases.
- **Table B** — genuine KB *extension*, not captioning, across 5 more Soybean diseases: refinements, relocations, and added differentials.
- Table B deltas are *proposed* refinements for human review before they overwrite canonical — flagged, not yet truth. The Bacterial Speck row in particular (added target-spot pattern) is exactly the kind of item a reviewer should adjudicate.